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PALEOMAGNETIC STUDY OF THRUST SHEET ROTATION DURING FORELAND IMPINGEMENT IN THE WYOMING-IDAHO OVERTHRUST BELT

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Abstract. Upper Jurassic and Lower Cretaceous strata in the Darby and Absaroka thrust sheets were sampled in order to paleomagnetically evaluate thrust sheet rotation in the Wyoming-Idaho overthrust belt. No significant rotations were found. Lack of rotation in these thrust sheets suggests that previously reported rotation of the Prospect thrust sheet was not transmitted to the older Darby and Absaroka thrust sheets, indicating that the Prospect thrust sheet only experienced local rotations in response to collision with the Teton-Gros Ventre uplift. The Upper Jurassic Stump Formation possesses a magnetization that passes the fold test and gives a paleomagnetic pole at 64°N, 170°E (K= 18, A_{95}= 15°), in good agreement with other published Upper Jurassic poles for cratonic North America. Samples from all Lower Cretaceous sites are believed to possess a synfolding remagnetization of chemical origin.

Introduction

Paleomagnetism was recognized early on as a technique that could document thrust sheet rotation in mobile belts [e.g., Kotasek and Krs, 1965]. After completing a seminar in paleomagnetism and the University of Michigan geology field course at Camp Davis, Wyoming, then-graduate student S. Tom Crough was the first to collect paleomagnetic samples in the Wyoming-Idaho overthrust belt near Camp Davis. Tom worked on this pilot collection for one semester, and his results were later incorporated in a publication [Grubbs and Van der Voo, 1976]. A decade later, we have returned to the area for a follow-up paleomagnetic study of thrust sheet rotations. We wish to dedicate this paper to the memory of Tom.

Traces of the major thrust sheets in the northern overthrust belt of western Wyoming and eastern Idaho strike N-S through most of Wyoming and Idaho, changing strike to a NW-SE orientation just south of Jackson, Wyoming (Figure 1). The origin of this curvature has been the subject of speculation and investigation, resulting in a variety of hypotheses for its origin. Irregularities in cratonic margins generated during rifting have been suggested as precursors to salients and reentrants developed during later orogenic activity [Thomas, 1983]. This process has been observed in the Appalachian orogenic belt [Rodgers, 1975; Rankin, 1976; Thomas, 1977] and has been suggested for the Wyoming-Idaho overthrust belt [Beutner, 1977]. A decrease in thrust sheet spacing, as well as an increase in thrust sheet imbrication in the vicinity of the Teton and Gros Ventre Ranges, led Horberg et al. [1949] to suggest that these foreland uplifts served as buttresses which impeded thrust sheet motion. This idea has now been widely accepted [Love, 1973; Dorr et al., 1977], but the consequences of such buttressing are still in question. A paleomagnetic study of Triassic strata in the Wyoming salient [Grubbs and Van der Voo, 1976] indicated that systematic rotation of thrust sheets developed in response to their impingement upon the buttresses. Portions of the thrust sheets in the vicinity of the basement uplifts rotated counterclockwise in the northern region, where they encountered the ancestral Teton-Gros Ventre block, and clockwise in the southern region, where their motion was impeded by the Game Hill high-angle reverse fault (Figure 1).

The rotated sites of Grubbs and Van der Voo [1976] are almost exclusively in the frontal Prospect thrust sheet (Figure 1), while those sites located on the Darby and Absaroka thrust sheets to the west show little or no rotation. In this study, Upper Jurassic and Lower Cretaceous strata on these westward thrust sheets were sampled to determine whether they have rotated also.

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Sampling and Geology

Seven to eleven oriented samples were taken per site from five formations using a portable hand-held rock drill. These samples were then cut into one or more specimens for analysis. Ten sites (5-14) were located in the Absaroka thrust sheet, three (2-4) in the Darby thrust sheet, and one (1) in the Bear thrust sheet, a splay off the frontal Prospect thrust sheet (Figure 1). Sites were chosen to yield a north-south distribution in order to adequately sample the change in structural trend of the Wyoming Salient.

Upper Jurassic sites (1-7) are in the Oxfordian Stump Formation [Imlay, 1952], which consists of glauconitic sandstones and siltstones, calcareous sandstones and oolitic limestones; they are the youngest Jurassic marine beds in this area (Figure 2). Stump sites are in beds of differing orientations, permitting a fold test.

The seven Cretaceous sites were collected from the Ephraim Conglomerate (13) and the Peterson Limestone (12 and 14), which are formations in the Gannett Group, the Smiths Formation (8 and 9), and the Wayan Formation (10 and 11). The Ephraim Conglomerate is the lowest formation in the Gannett Group (Figure 2), a sequence that overlies the Stump Formation and that is believed to span the Jurassic-Cretaceous boundary [Eyer, 1969]. Our samples are red siltstones, common in this formation. The Peterson Limestone is lacustrine, and is also part of the nonmarine Gannett Group [Glass and Wilkinson, 1980]. Lacustrine floras and faunas in the Peterson Limestone indicate that it is Aptian in age [Eyer, 1969]. The Smiths Formation overlies the Gannett Group in westernmost Wyoming and southeastern Idaho and consists of dark shales and purple to green-grey, fine-grained quartzitic sandstones or siltstones in the sampling area. Imlay [1952] assigned an Albian age to the lower Bear River Formation, the eastern lateral equivalent of the Smiths Formation (Figure 2). The Wayan Formation conformably overlies the Smiths Formation. It also is assigned to the Albian stage on the basis of a nonmarine flora [Schmitt and Moran, 1982]. Both a coarse-grained sandstone unit (11) and a red siltstone unit (10) were sampled.

Five of the seven Lower Cretaceous sites (8-12) are within a series of anticlines and synclines that are in

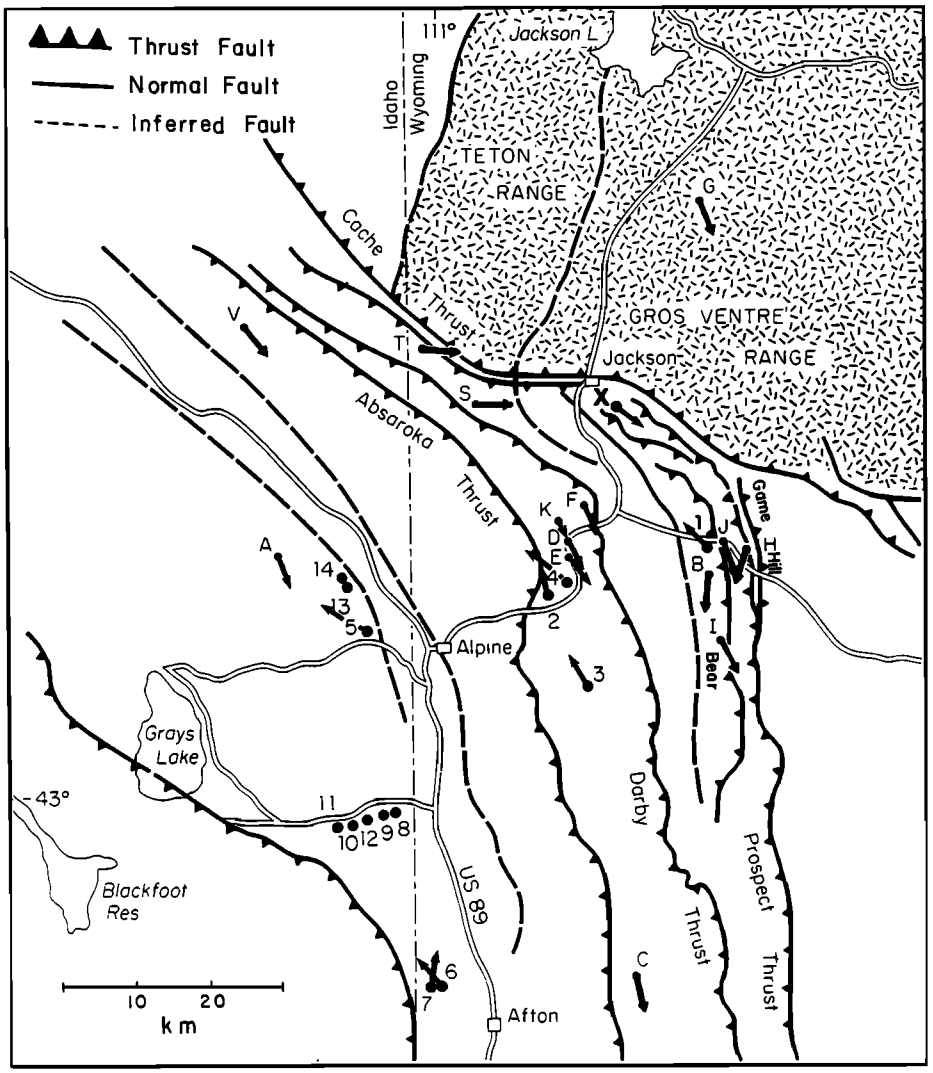


Fig. 1. Map of the Wyoming-Idaho overthrust belt. Numbered sites (1-14) are Jurassic and Cretaceous sampling localities from this study; lettered sites are Triassic sampling localities from Grubbs and Van der Voo [1976]. Arrows represent the mean-site declinations; directions from lettered sites are reversed relative to those from numbered sites. Site G lies in the stable foreland, and its declination is used as the Triassic reference declination. A declination of 330°, as derived from the Upper Morrison pole [Steiner and Helsley, 1975], was used as the reference for the Jurassic Stump sites (not shown). Thrust sheets wrap around the ancestral Teton-Gros Ventre block (patterned region) giving rise to the Wyoming salient (concave toward the craton). Note that systematic deviations of mean-site declinations from the reference are apparent in the Prospect thrust system.

structural continuity with Stump sites 6 and 7; these structures have been disrupted in the area of Figure 1 by later high-angle faults [Oriel and Platt, 1980].

Paleomagnetic Methods and Results

All characteristic directions were determined after specimen demagnetization. Specimens were subjected to stepwise thermal demagnetization with a Schonstedt TSD-1 furnace, alternating field (AF) demagnetization in a Schonstedt GSD-1 specimen demagnetizer, or chemical leaching using 10 normal hydrochloric acid. Specimens from the same sample possess similar paleomagnetic directions and rock magnetic properties. Site-mean directions were calculated by first averaging specimen directions for each sample and then using sample directions to obtain site-mean directions. For this reason, the word sample rather than specimen is used in the following except when referring to demagnetization diagrams.

Jurassic Stump Sites (1-7)

Thermal and alternating field demagnetization were used to isolate characteristic directions in the Stump Formation samples. Both techniques yielded similar directions, but alternating field treatment did not always result in a decay of the magnetization sufficient to reach the origin (Figure 3b). For this reason, most samples were thermally demagnetized. Demagnetization diagrams for three specimens from different sites (Figure 3) illustrate characteristic behavior during

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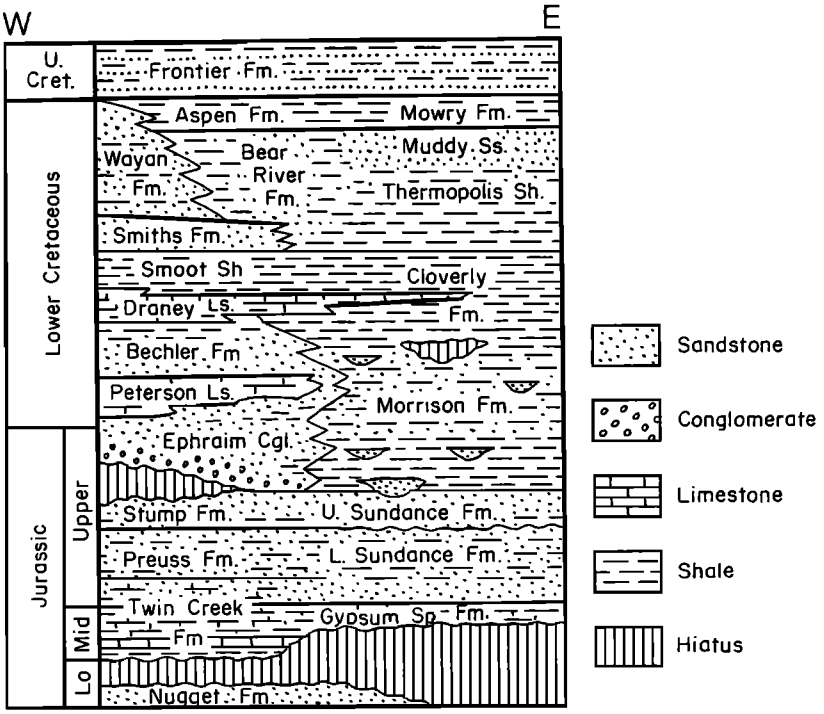


Fig. 2. Generalized stratigraphic column of the Upper Mesozoic rocks in western Wyoming and surrounding regions.

thermal and AF treatment. A specimen from site 1 (Figure 3b) was first subjected to alternating field demagnetization up to 100 mT and then subsequently thermally treated to 675°C. After removal of a soft magnetization in fields below 20 mT, the decay of magnetization followed a linear trajectory to the origin. Thermal treatment of a specimen from site 4 (Figure 3c) and AF treatment of a specimen from site 6 (Figure 3a) removed only one direction, resulting in an apparently univectorial decay of magnetization toward the origin.

Acquisition of isothermal remanent magnetization (IRM) by samples from sites 1 and 4 reveals that saturation is not reached until fields well above 3 T, suggesting the presence of hematite. Moreover, the AF and thermal demagnetization behavior of samples from these sites as well as from site 2, strongly suggests that hematite is the dominant magnetic carrier in these rocks. The magnetization in these samples typically persists up to 675°C and is only partially removed in fields up to 100 mT. Samples from the remaining four sites (3,5,6, and 7) show a markedly different demagnetization behavior. The magnetization of these samples decays significantly in alternating fields of 90 mT, and they have blocking temperatures below 575°C, suggesting only the presence of (titano-?)magnetite. Sample directions for all Stump Formation sites were determined by vector subtraction with the exception of samples from site 2; they retained a viscous remanent magnetization parallel to the present field direction even after treatment with alternating fields up to 70 mT and were, in general, unstable during thermal cleaning. After AF treatment, a common direction believed to be the characteristic direction was isolated in most samples by the convergence of remagnetization circles [Halls, 1976].

Site-mean directions both before and after correction for structural tilt are listed in Table 1 and are shown in Figure 4. It is apparent that the dispersion of site-mean directions is dramatically reduced when directions are rotated into their tilt-corrected configuration. This increase in clustering upon tilt correction results in a positive fold test that is significant at the 99% confidence level [McElhinny, 1964], restricting the age of magnetization to be pre-Sevier (pre-Clarkforkian) deformation. The paleomagnetic pole position calculated for these seven sites of the Stump Formation is located at 64°N and 170°E.

Cretaceous Sites (8-14)

Samples from the seven Lower Cretaceous sites vary in their magnetic mineralogy and therefore in their magnetic behavior. Samples from the Smiths sites (8 and 9) and the Peterson sites (12 and 14) contain (titano-?)magnetite as their dominant magnetic carrier. This is suggested by their IRM acquisition curves, low coercivities, and for site 8, a blocking temperature below 500° C (Figure 5a). Because of the poor response of the Peterson Limestone to thermal demagnetization, most samples from sites 12 and 14 were cleaned using AF demagnetization. All samples from the Ephraim Conglomerate (site 13) were thermally demagnetized and display a univectorial decay to the origin after removal of a soft component by 400°C (Figure 5b). High blocking temperatures and results of IRM experiments indicate hematite as the magnetic carrier. Saturation of IRM in magnetizing fields in excess of 10 T for samples from the Wayan sites (10 and 11) indicates the presence of hematite, confirmed by blocking temperatures up to 615°C. Samples from the red siltstone unit of the Wayan Formation (site 10) were most successfully demagnetized using chemical leaching, while thermal treatment yielded the best results for the sandstone unit sampled at site 11.

The weak intensity (between 1 and 2 x 10^{-4} A m^{-1}) of samples from Peterson site 14 caused demagnetization

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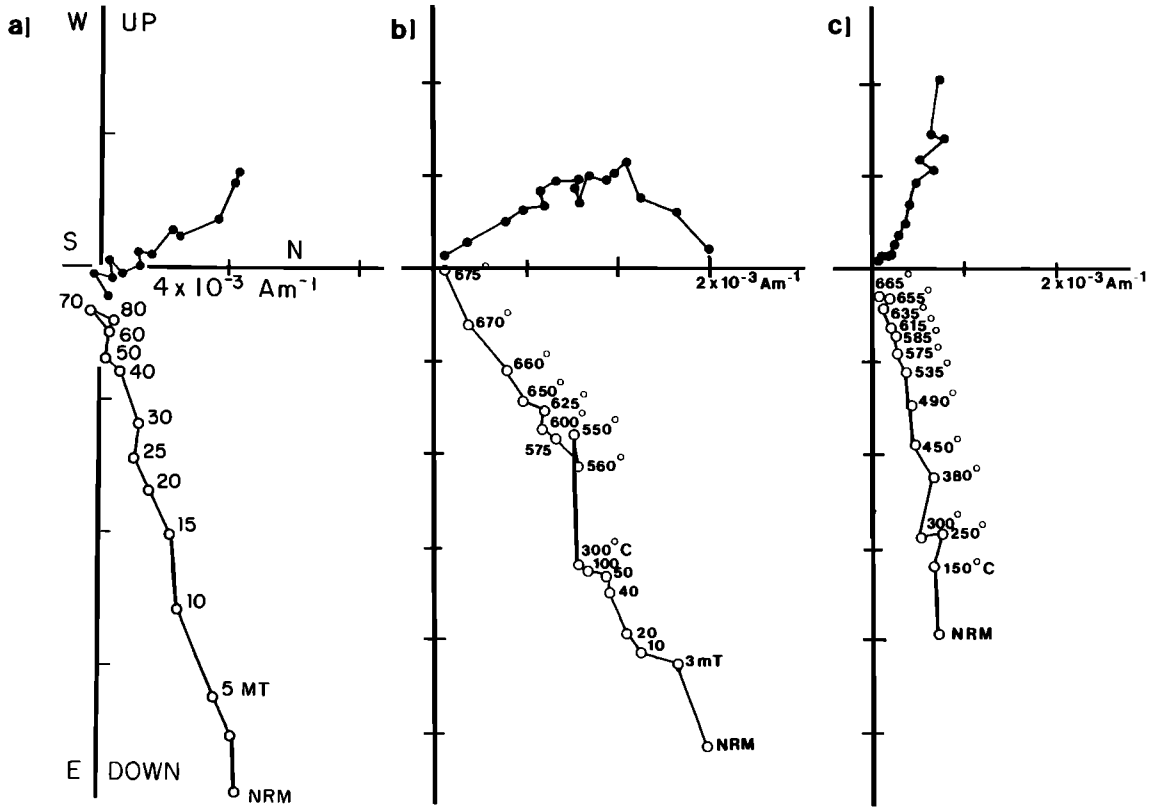


Fig. 3. Orthogonal demagnetization diagrams for three Stump specimens. (a) The response to stepwise AF demagnetization of a specimen from site 6, (b) AF and thermal demagnetization of a specimen from site 1, (c) thermal demagnetization of a specimen from site 4. Solid circles represent projections on the horizontal plane, open circles on the N-S vertical plane. Coordinate axes are similar for all three diagrams. Vector directions have been corrected for structural tilt.

diagrams to become too noisy at higher fields to accurately determine directions using vector subtraction. Great-circle analysis [Halls, 1976] of trajectories at site 14 allowed an estimate of the pole to the plane containing all great-circle poles and was successful at determining a site-mean direction that was consistent with directions obtained using vector subtraction where possible. For all other Cretaceous sites, sample directions were determined by vector subtraction.

Site-mean directions from the seven Lower Cretaceous sites (Table 2) cluster poorly both before and after tilt correction (Figure 6), yet show an increase in clustering at an intermediate position along the path they follow during structural correction. If the Cretaceous site-mean directions are untilted about their strike for only one-half of their dip angle (i.e., unfolded 50%), the resulting directions are in good agreement. Two explanations are consistent with these observations: (1) simultaneous removal of two components in a multicomponent magnetization, giving rise to directional deviation and apparent crossing over of site-mean directions during structural correction because of undetected but incomplete cleaning, and (2) a synfolding remagnetization. The possibility that the dispersion in site-mean directions is caused by the rotation of individual sites about vertical axes has been rejected, because such rotations would not produce the observed dispersion in inclinations. We favor a synfolding remagnetization for the following reasons.

1. Decay of the last 20% of the magnetization in samples from sites 8 and 13 is linear (Figure 5), suggesting only one component of magnetization. However, the mean directions from these sites still pass each other during structural correction for the dip of the beds (Figure 6).

2. After tilt correction, the mean direction for site 11 moves away from the other site-mean directions (Figure 6). The demagnetization diagrams for specimens from this site show no evidence of incomplete removal of a component of magnetization capable of causing the directions to move toward the other Lower Cretaceous site-mean directions.

In order to test this synfolding hypothesis, the values of k at intermediate unfoldings were compared with the value of k both before and after tilt correction. If both these k ratios pass the F-test for the given number of sites, then a synfolding magnetization would be significant to the confidence level tested. In other words, the dispersion of site-mean directions at an intermediate percentage of unfolding must be significantly better than the dispersion both before and after tilt correction. The assumption that displacement of both fold limbs is commensurate and contemporaneous is implicit in this test.

A graph of the k values for different percentages of unfolding for these seven sites (Table 2) reveals a maximum at about 60% unfolding (Figure 7). The k value required for a synfolding magnetization significant at the 95% confidence level corresponds to unfolding between 55% and 70%. This indicates that the most probable timing for the acquisition of magnetization is during the second quarter of the folding episode.

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TABLE 1. Summary of Paleomagnetic Results From the Jurassic Stump Formation

Site	Formation	N/N_o	K	α_{95}	Characteristic Direction D/I Before Tilt Correction	After Tilt Correction
1	Js	8/9	43	9	278/11	310/56
2	Js	4/7			270/15	343/60
3	Js	8/8	16	14	354/43	329/47
4	Js	9/9	105	5	10/47	310/57
5	Js	6/8	22	13	334/40	298/61
6	Js	8/9	168	4	32/49	318/73
7	Js	9/9	32	9	276/31	7/63
Mean					316/44 $k=4$, $\alpha_{95}=35$	324/61 $k=35$, $\alpha_{95}=10$ pole 64°N 170°E $K=18$, $A_{95}=15$

N/N_o is the number of sample directions used in the calculation of a site-mean direction compared with the number of samples demagnetized; k, K, α_{95}, and A_{95} are Fisher statistical parameters; D/I is declination and inclination, and Js is Jurassic Stump Formation.

Discussion

Although the seven sites in the Stump Formation are distributed within three different thrust sheets, we demonstrate below that their site-mean directions show no systematic deviations in declination that can be correlated with their location or with local structural trends. Thus the mean of all seven sites may be used to evaluate the hypothesis of rotation of the thrust belt as a whole with respect to the craton. The paleomagnetic pole position calculated for these seven sites of the Stump Formation is located at 64°N and 170°E. Since a reference Stump site collected in the (presumed) stable foreland yielded uninterpretable results, the pole from this study must be compared with other published Middle and Upper Jurassic poles in order to evaluate the possibility of rotation. If rotation about a vertical axis had taken place between the thrust belt in the vicinity of the sampling area and cratonic North America, the pole would be expected to fall along the arc drawn through the Stump pole in Figure 8. A clockwise rotation of the sites relative to North America would displace the pole to the northwest along the arc, whereas a counterclockwise rotation would result in a southeasterly displacement of the pole. The pole calculated in this study lies within the margin of error for the path of poles from the Early Jurassic through the Late Cretaceous (Figure 8). It is in particularly good agreement with the pole determined from the Upper Morrison Formation which is thought to be Kimmeridgian to Tithonian in age [Steiner and Helsley, 1975]. Since the pole from the Stump Formation lies on the path of Jurassic through Cretaceous polar motion for North America, no significant net rotation about a vertical axis has taken place between the thrust belt in the vicinity of the sampling area and cratonic North America.

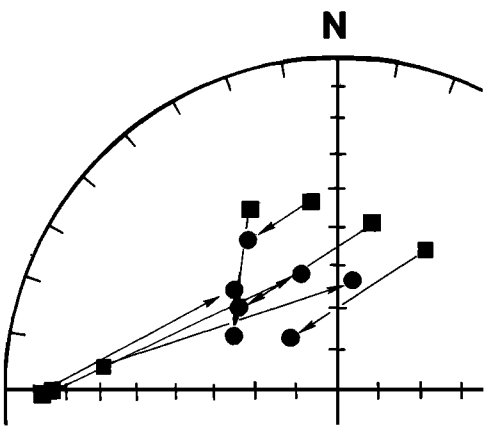


Fig. 4. Equal area plot of site-mean directions both before (squares) and after (circles) correction for structural tilt for all Jurassic Stump sites. Closed symbols represent positive (down) inclinations.

As mentioned earlier, we interpret the magnetizations of the seven Cretaceous sites to be of synfolding age. The requirement that the magnetization was acquired during folding restricts its age to be older than the last episode of deformation (63-55 Ma). Movement on the Absaroka thrust fault (all seven Cretaceous sites are located within the Absaroka thrust sheet) is well-dated as beginning in the Santonian and continuing through the Maestrichtian, thus ranging from 80 to 65 m.y. B.P. [Wiltschko and Dorr, 1983]. Table 2 lists the site-mean directions and the overall paleomagnetic poles at 0%, 100%, and 60% unfolding. We believe that the 60% pole at 69°N, 204°E best represents the paleofield at the time of magnetization acquisition. A comparison of this pole with the polar wander path for North America [e.g., Harrison and Lindh, 1982] is consistent with a Late Cretaceous age of magnetization. These seven Lower Cretaceous sites need not therefore be eliminated as indicators of possible rotations. To determine whether or not remagnetized directions would have been affected by rotations, the timing of both events must be investigated further. If we assume that any possible rotation of the Absaroka sheet took place during thrusting, rotation might have preceded

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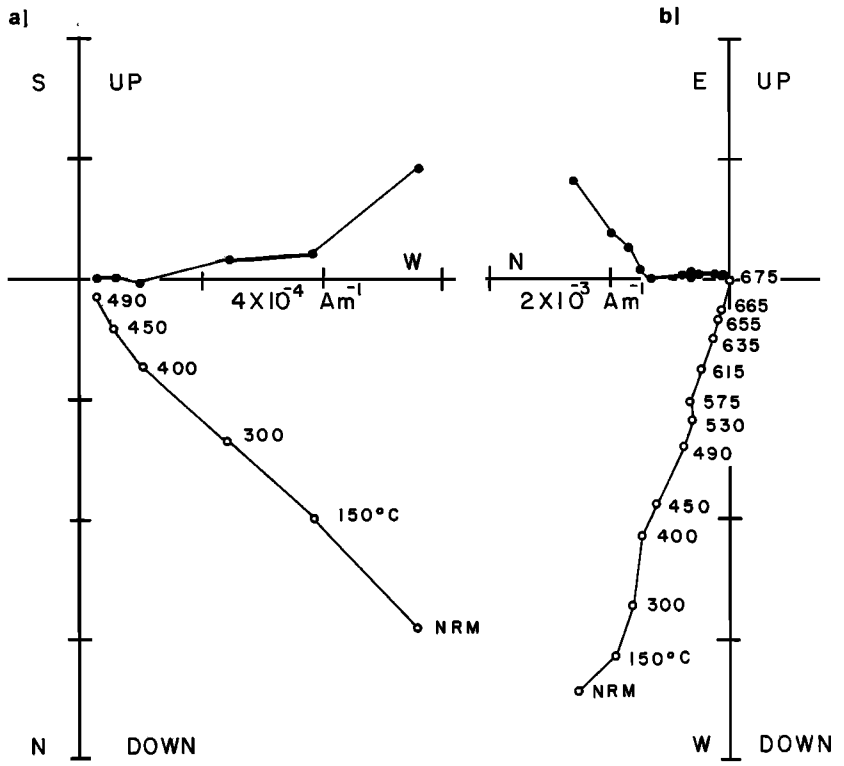


Fig. 5. Orthogonal demagnetization diagrams of stepwise thermal demagnetization of a specimen (a) from the Smiths Formation, site, 8 and (b) from the Ephraim Conglomerate, site 13. Solid circles represent projections on the horizontal plane, open circles on the vertical plane. Note that the coordinate axes differ for the two diagrams. Vector directions have been corrected for structural tilt.

remagnetization, making recognition of the rotation impossible. However, movement on the younger thrust faults (Darby and Prospect), which lie beneath the Absaroka thrust sheet, would have translated any rotations they experienced to the Absaroka thrust sheet. Since movement on the Darby and Prospect thrust sheets was the last episode of Sevier deformation, any rotations occurring in these thrust sheets should be detectable in the remagnetized directions from the Absaroka thrust sheet. No such rotations are observed.

Since the paleomagnetic data available for evaluation of possible rotations in the Wyoming-Idaho overthrust

TABLE 2. Summary of Paleomagnetic Results From the Cretaceous Sites

Site	N/N_o	Formation	k	α_{95}	Characteristic Direction D/I: Before Tilt Correction	After Tilt Correction	60% Correction
8	8/8	Ks	113	5	47/50	269/46	304/73
9	7/7	Ks	58	8	51/48	251/69	14/83
10	7/9	Kw	39	10	356/46	304/73	339/65
11	9/10	Kw	14	15	326/58	22/48	5/57
12	9/11	Kp	15	14	55/39	298/56	357/70
13	7/7	Ke	70	7	275/47	353/66	309/66
14	5/6	Kp			267/54	327/64	299/64
Mean					0/65	313/68	333/71
					k=5, α_{95}=30	k=12, α_{95}=18	k=37, α_{95}=10
Mean poles					86°N, 249°E	58°N, 190°E	69°N, 204°E

N/N_o is the number of sample directions used in the calculation of a site-mean direction compared with the number of samples demagnetized; k, K, α_{95}, and A_{95} are Fisher statistical parameters; D/I is declination and inclination; Ks is Smiths Formation, Kw is Wayan Formation, Ke is Ephraim Conglomerate, and Kp is Peterson Limestone.

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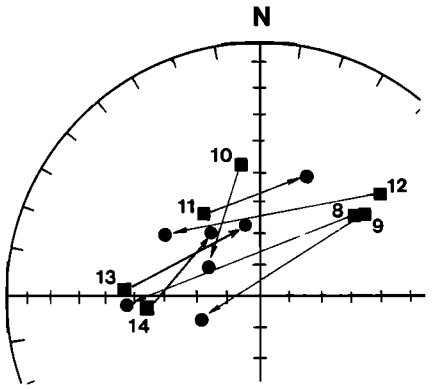


Fig. 6. Equal area plot of site-mean directions before (squares) and after (circles) structural correction for the Lower Cretaceous sites. Sites 8 and 9 are from the Smiths Formation, sites 10 and 11 are from the Wayan Formation, sites 12 and 14 are from the Peterson Limestone, and site 13 is from the Ephraim Conglomerate. Solid symbols represent positive (down) inclinations.

belt span the Triassic through the Cretaceous, a graphic representation of the data capable of removing any age dependence is desirable. In such a technique, the deviation of site-mean declinations from appropriate reference declinations is plotted with respect to strike deviations from a chosen standard. A reference declination of 158° reported by Grubbs and Van der Voo [1976] was used as a standard with which to compare their Triassic site-mean declinations. The Upper Morrison pole was chosen as a contemporaneous reference paleopole with which to compare the Stump site-mean declinations. The age of magnetization residing in the Cretaceous samples is assumed to be Late Cretaceous (Santonian-Maestrichtian), and a reference declination of 331° has been used as derived from the Elkhorn Mountain volcanics paleopole [Hanna, 1973].

The results of this analysis show two distinct trends. Sites from the Prospect and related Bear thrust sheets show a significant correlation between the declination deviation from the reference and the deviation of the thrust trace from the chosen standard (Figure 9a). The resulting product mean correlation coefficient for the two parameters is 0.82, and the 95% confidence limit for the slope of the best fit line through the data contains a slope of +1 (i.e., a one-to-one correspondence between variables). A similar correlation does not exist for the sites in the Darby and Absaroka thrust sheets (Figure 9b), where the correlation coefficient is 0.23. A strike parallel to the Triassic reference declination direction was chosen for convenience as the standard for the thrust traces; however, if any other orientation had been used, the correlation would remain the same. This analysis strongly suggests that only the eastern part of the Prospect thrust sheet has experienced any significant rotation during impingement upon the foreland uplifts. The rotation of the Prospect thrust sheet is believed to be an "edge effect" that probably has not displaced the portion of the thrust sheet that lies beneath the older thrust sheets.

Thrust sheet deformation paths derived from fracture patterns and calcite twin compression directions (A. Kopania and D. V. Wiltschko, personal communication, 1983) begin to deviate significantly from linear trajectories as the Prospect thrust sheet is approached. These researchers suggest thrust sheet imbrication, transverse faulting, and oblique folding as possible mechanisms to accommodate their inferred rotations. As movement of the Prospect thrust became progressively impeded by the uplifted ancestral Teton-Gros Ventre block, its displacement must have been transferred from the main fault to the associated structures suggested by Kopania and Wiltschko.

Displacement transfer from thrust faults to associated structures has been modeled by Gardner and Spang [1973] using clay analogues. They find that shear zones, as precursors to transverse tear faults, develop at the terminus of a fault. Such tear faults in the Wyoming-Idaho overthrust belt are readily observed in the surface geology [Dorr et al., 1977]. Dixon [1982] has derived displacement curves for the various thrust systems in the Wyoming-Idaho overthrust belt through systematic restoration of thrust sheet geometries to their prethrusting configuration. These displacement curves mimic the shape of the thrust traces, indicating that differential displacement has been active in this thrust belt and is primarily responsible for its curved nature. Dixon also notes a spatial coincidence of ob-

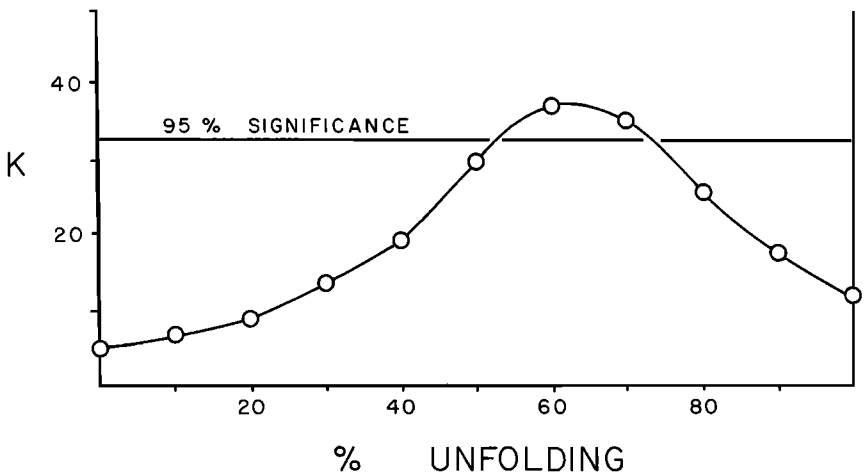


Fig. 7. Graph of the Fisher precision parameter k against the percentage of unfolding of the beds for sites 8-14. Here 100% unfolding represents flat lying beds or beds that have been tilt corrected. The 95% confidence level is marked and was calculated with respect to the k value at 100% unfolding, since this value is greater than the k value at 0% unfolding. The k values plotting above this line pass a synfolding fold test at the 95% confidence level.

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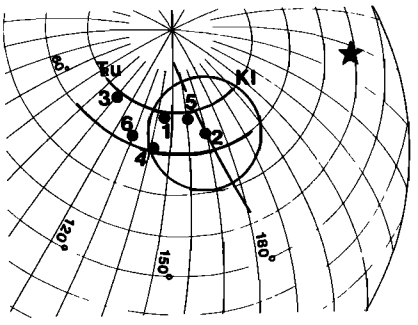


Fig. 8. Orthographic projection showing the location of mean Upper Triassic (T_Ru) and Lower Cretaceous (Kl) poles and individual Jurassic poles: 1, Twin Creek [McCabe et al., 1982]; 2, Stump (this study); 3, Summerville [Steiner, 1978]; 4,Lower Morrison, 5, Upper Morrison [Steiner and Helsley, 1975]; 6, Canelo Hills [Kluth et al., 1982]. The circle represents the 95% confidence limit for the Stump pole. The arc represents the path the pole would be expected to fall on, if rotation about a local vertical axis with respect to cratonic North America had taken place. The star marks the sampling locality.

lique folds with maximum and minimum displacement locations, suggesting these as a mechanism for displacement transfer.

In their clay models, Gardner and Spang [1973] also observed a "fanning effect" caused by extension parallel to the leading edge of the thrust; here, the local and regional transport directions diverge as the fault terminus is approached. Stress trajectories run parallel to the regional transport direction only near the midpoint of the fault, and fan away from this direction toward the ends of the fault.

The observations of oblique folds, the presence of tear faults, the fanning effect, and the increase in thrust sheet imbrication as the foreland buttresses are approached, all indicate the transfer of displacement from the Prospect thrust sheet to secondary structures. In these ways, rotation of the Prospect thrust sheet in contact with the buttresses was accomplished while the older westward thrust sheets experienced neither displacement nor rotation.

The origin of the remagnetization residing in the Cretaceous sites is problematic. A thermal remagnetization must be rejected, since blocking temperatures of samples from these sites far exceed the thermal conditions in the overthrust belt during Sevier deformation (Jurassic to Paleocene). Therefore a chemical remagnetization is suggested. Because these sites are all within the Absaroka thrust sheet, which has experienced the longest and most continuous movement compared with the other thrust faults in the sampling area [Blackstone, 1977], and because faults often act as fluid conduits, the long history of movement on the Absaroka fault could generate the steady supply of fluids required for a chemical remagnetization.

Conclusions

A total of 20 Triassic through Cretaceous sites from the Darby and Absaroka thrust sheets at various locations in the Wyoming Salient reveals that these two thrust sheets have undergone no systematic rotation with respect to the North American craton. However, eight sites collected from the Prospect thrust system do display rotations that correlate with the trends of the thrust traces. These observations suggest that only the Prospect thrust sheet underwent local rotations in

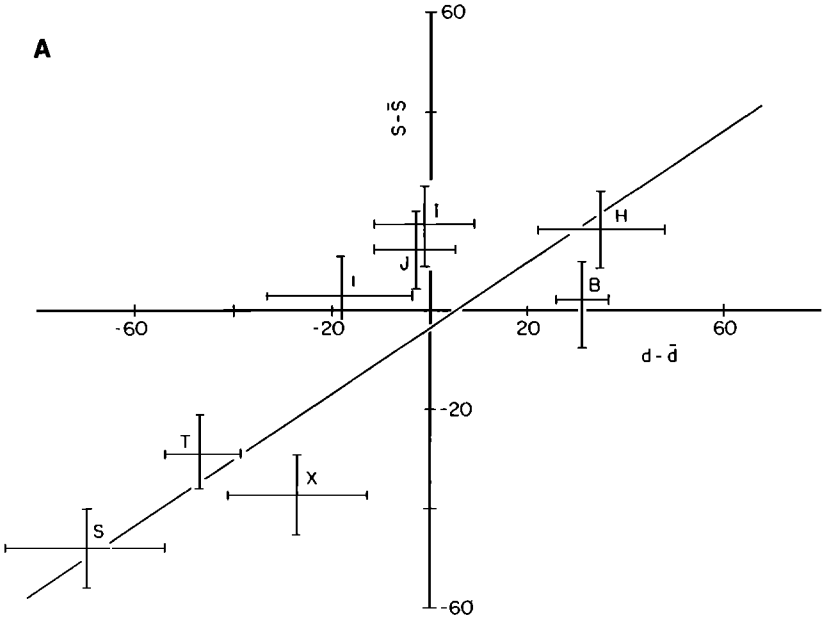


Fig. 9a. Graph of site-mean declination deviations from an appropriate reference value, against thrust sheet strike deviations from a reference value, using the Triassic data of Grubbs and Van der Voo [1976] as well as the Jurassic and Cretaceous data from this study for sites from the Prospect thrust system. A strong correlation between these two parameters is evident giving a correlation coefficient of 0.82. The error in the declinations was calculated by dividing the cone of confidence of the site-mean directions by the cosine of the site-mean inclinations. The declinations of the sites that plot with the same strike were averaged.

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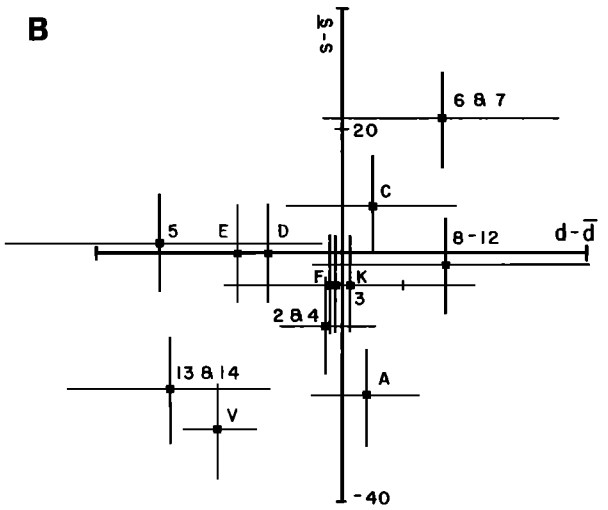


Fig. 9b. Same as Figure 9a for sites from the Darby and Absaroka thrust sheets. Here no correlation is evident with a correlation coefficient of 0.23.

response to its collision with the Teton-Gros Ventre uplift and the Game Hill high-angle reverse fault. These rotations were not transmitted to the older thrust sheets, westward from the craton, but were transferred to secondary structures.

The Upper Jurassic Stump Formation of western Wyoming and southeastern Idaho has been found to possess a stable prefolding magnetization, yielding a paleomagnetic pole at 64°N and 170°E (seven sites, D=324°, I=+61°, α_{95}=10.4°, k=35). This paleopole is in good agreement with a Late Jurassic pole determined from the cratonic Upper Morrison Formation.

A synfolding remagnetization is believed to have affected the Lower Cretaceous rock units sampled from the Absaroka thrust sheet. This remagnetization is thought to be of diagenetic origin, acquired during the second quarter of the folding episode.

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